COMP3010 Machine Learning

REPORT

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1 Introduction

With the problem assignment at hand to accurately predict the peak pressure caused by Boiling Liquid Expanding Vapor Explosions (BLEVEs). I was assigned to research the various machine learning models to aid me in finding the suitable on to address the issue. Firstly, I had to face the challenges that the data set came with before I could pass the data set to the models. This report outlines the journey that I took, starting from data cleaning of the dataset, processing it, model selection, hyperparameter tuning, and lastly the final predictions on the test set based on model ensemble.

Kaggle Name

Sycolee

2 Data cleaning

After careful and insightful analysing of the train.csv files I found the following issues that would clutter with the model.

2.1 Issues

1. The data had some instances where the entry was missing some features.
2. The data frame given had some outliers in instance features.
3. The data also had duplicate instances in the frame.
4. There were also features in the data that had no correlation to the problem statement given.
5. The data frame did not follow one data type throughout the all the features.
6. The features were not on the same scale.

2.2 Solutions

1. For missing entries in the data file, I utilised imputation through a function. The function will use the `SimpleImputer` class from the scikit-learn library to impute missing values for numeric and categorical columns. For the numeric feature the function will adapt the 'mean' method for imputation and for catergorical features the 'most_frequent' method was used for imputation.
2. For the outliers I used a function to detect and remove them from the dataset. The function was implied by setting qualitative bounds that suited the data frame for the numeric feature and removed from the set. For the categorical columns I derived it on the uniqueness of an object found in the data frame if it did not fall under a common object values it would be removed.
3. For duplicated in the data a function was implemented, to check if there was any duplicated in the frame by simple duplicated. Sum () method and remove them if any duplicates are found.
4. A cost function was done to determine the threshold that was suitable to put on the target variable as correlation of other features to the target. Using a function, I was able to select the appropriate columns that had correlation to the data frame.
5. I used one-hot encoding on the features of the data frame because I identified that there, we categorial columns that would affect the model if not encoded properly to fit the reset of the data.
6. Used Normalisation as a function that was used to scale the features in the dataset using the normalization technique.

3 Data processing

After the issues in in section 2.1 were solved using various data processing techniques the dataset was left clean and left for further processes that would aid the model efficiency.

1. Feature engineering

   This required me to creating new features from the existing features in the dataset which would help in the model acuracy. Some of the features that were generated in the function include: volume, volume to surface area ratio, width to length ratio. These features had high correlation to the target variable hence aiming to a successful model.

2. Feature Normalisation

   To ensure that all the feature on the same scale I implemented a function that would be used to scale the features in the dataset using the normalization technique. The function used was the `MinMaxScaler` class from the scikit-learn library to scale the features in the dataset.

3. Data type conversion

   To ensure that the whole dataset would follow the same numerice data type I unsed a function that converted all the features to floats to ensure that the data is accurate to the model.

4 Model selection.

In investigating and selecting models for predicting the peak pressure caused by BLEVEs, I explored a variety of regression models to ensure diversity and relevance to the problem at hand. Here's a discussion on each model investigated:

4.1 Linear Regression Model

Linear regression is a fundamental modelling technique that assumes a linear relationship between the independent variables and the target variable. Given the simplicity of its assumptions, I decided to include a Linear Regression model in my investigation. However, due to the complexity of the problem and the potential non-linearity of the relationships between the features and the target variable, I ultimately rejected this model as it may not capture the underlying patterns well.

4.2 SVR Model

Support Vector Regression (SVR) is a powerful regression technique that can capture complex relationships between variables, especially in high-dimensional spaces. I chose SVR for its flexibility in handling non-linear relationships and its ability to incorporate kernel functions to transform the data into higher dimensions. This model is relevant to our problem as it can potentially capture the complex interactions between the various features and the target variable, making it a suitable candidate for prediction.

4.3 Random Forest Model

Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting. I selected Random Forest for its robustness to noise, ability to handle non-linear relationships, and feature importance estimation. Given the diversity of features in the dataset and the potential interactions between them, Random Forest is well-suited for capturing complex patterns and providing accurate predictions.

4.4 XGBoost Model

XGBoost is a popular gradient boosting library known for its efficiency, scalability, and performance. I included XGBoost in my investigation for its ability to handle large datasets, feature importance analysis, and regularization techniques to prevent overfitting. With its ensemble of weak learners, XGBoost can effectively capture intricate relationships in the data and provide accurate predictions.

Overall, I selected SVR, Random Forest, and XGBoost models for their diversity in modelling approaches and their relevance to the problem of predicting peak pressure caused by BLEVEs. One of the major reasons as well was that SVR and Random Forest are good together in model ensemble, so I opted to use these in my assignment.

5 Hyperparameter tuning.

For the Hyper tunning I used grid search for my models that I implemented on the dataset and below is an explanation on how the result was achieved:

1. SVR model

   I employed grid search for hyperparameter tuning. The hyperparameters tuned were 'C' and 'epsilon'. I considered a grid of values for each hyperparameter: [0.1, 1, 10, 100] for 'C' and [0.1, 0.2, 0.3, 0.4] for 'epsilon'. After tuning, the optimal values determined were 'C': 10 and 'epsilon': 0.1.

2. Random Forest model,

   I also used grid search for hyperparameter tuning. The hyperparameters tuned were 'number of estimators (n_estimators)' and 'maximum depth of the trees (maximum depth)'. I considered a grid of values for each hyperparameter: [50, 100, 200, 300] for 'n_estimators' and [5, 10, 15, 20] for 'maximum depth'. After tuning, the optimal values determined were 'n_estimators': 300 and 'maximum depth': 10.

3. XGBoost model

   I employed grid search for hyperparameter tuning. The hyperparameters tuned were 'number of estimators (n_estimators)' and 'maximum depth of the trees (maximum depth)', with the same grid of values considered as for the Random Forest model. After tuning, the optimal values determined were 'n_estimators': 200 and 'maximum depth': 5."

6 Prediction

Explain how the final predictions on the test set were derived. This might include the use of a single model, a combination of models, or an ensemble method.

For the final prediction file, I managed to have a csv that had the predicted Target pressure with iterations will 3200 prediction file sample shown below:

	I...	Target Pressure (bar) (2)
1	ID	Target Pressure (bar)
2	0	0.22277132
3	1	0.43194613
4	2	0.32764596
5	3	0.26864684
6	4	0.19803983
7	5	0.12416377
8	6	0.093866214
9	7	0.1285154

In this approach, I combined the predictions from both SVR and Random Forest models using a Voting Regressor. The Voting Regressor averages the predictions from both models, leveraging the strengths of each individual model. After training the Voting Regressor on the training data, I evaluated its performance on the test set using evaluation metrics such as Mean Absolute Percentage Error (MAPE) and R-squared (R2).

Finally, I derived the final predictions on the test set and generated model graphs to visualize the predictions compared to the actual target values.

7 Self-reflection and Conclusion

his section is more open-ended and personal. Reflect on your overall understanding and feelings about the project. Discuss any difficulties encountered, lessons learned, and how you might approach the project differently if given another. chance. Feel free to include any other relevant thoughts or insights.

At the end of this project, I have understood a lot in terms of Models that are implemented in machine learning model. The implementation was more hands on to what I have been studying in class in for terminologies that I explore on my own. I thought the task of implementation the model was going to be easy as I am passionate about data and machine learning as whole, but I was caught off guard with the task presented to me.

I found the assignment to be very challenging as there was so many things that you had to figure out on your own assessment this was not easy task as the details of these modifications were very minute and hard to find a comprehend. Spend most of time trying to clean data and dealing which got a little bit grating with time. The assignment did not only require lot of quantity time but also quality time to be effective in finding what suits the model the most to come up with predictions that were close to the testing.

I was very excited with the competition that was presented on Kaggle wanted to show the best version of predictions that I could come up with, to my astonishment was nearly last in my MAPE score. I wish the positioning of this did not demotivate me in continuing to progress my model but later came on and had no time to improve it. I really enjoyed this adventure as it was just a highlight to what I will be doing in the future.